

Math Thermo
class #11
02/24/2026

Continuous Phase Space

Again we will only consider the case of N non-interacting particles. The energy of each particle is assumed to be a function of the form

$$(11.1) \quad u(\vec{x}, \vec{v}) = \frac{1}{2} m |\vec{v}|^2 + \psi(\vec{x}),$$

where $\vec{x} \in \Omega$ is the position of the particle in the domain Ω and $\vec{v} \in \mathbb{R}^3$ is its velocity vector.

The energy of any one particle is independent of the position and velocity of the other particles in the system.

m is the mass of the particle.

ψ is a potential energy, which depends upon only the position of the particles.

The term $\frac{1}{2} m |\vec{v}|^2$ is called the kinetic energy.

The expression (11.1) is an idealization because it neglects energetic interactions between particles.

If N is large, it is too hard to track particles individually. We use a density instead.

Definition (11.1): let $\Omega \subseteq \mathbb{R}^3$ be a simply - connected domain with volume

$$V = \int_{\Omega} d\vec{x}.$$

The number density of particles is a function

$$f: \Omega \times \mathbb{R}^3 \rightarrow [0, \infty)$$

with the property

$$(11.2) \quad \int_{\Omega} \int_{\mathbb{R}^3} f(\vec{x}, \vec{v}) d\vec{v} d\vec{x} = N > 0$$

where N is the total number of particles in Ω .
The function

$$p: \Omega \times \mathbb{R}^3 \rightarrow [0, \infty)$$

defined as

$$p(\vec{x}, \vec{v}) := \frac{f(\vec{x}, \vec{v})}{N}$$

is called the particle probability function. of course

$$(11.3) \quad \int_{\Omega} \int_{\mathbb{R}^3} p(\vec{x}, \vec{v}) d\vec{v} d\vec{x} = 1.$$

We assume that there is a fixed and finite amount of energy in the system

$$(11.4) \quad U = \iint_{\Omega \times \mathbb{R}^3} f(\vec{x}, \vec{v}) u(\vec{x}, \vec{v}) d\vec{v} d\vec{x}$$

The set $Q = \Omega \times \mathbb{R}^3$ is called phase space. It is of course in $\mathbb{R}^6$!

Suppose $A \subseteq Q$. The number of particles in A is

$$N \geq N_A := \iint_A f(\vec{x}, \vec{v}) d\vec{v} d\vec{x}.$$

The energy of the particles in that set is denoted

$$U \geq U_A := \iint_A f(\vec{x}, \vec{v}) u(\vec{x}, \vec{v}) d\vec{v} d\vec{x}.$$

Definition (11.2): The average energy (per particle) is defined as

$$\bar{U} = \iint_Q f(\vec{x}, \vec{v}) u(\vec{x}, \vec{v}) d\vec{v} d\vec{x}.$$

Of course, it must be that

$$(11.5) \quad U = N \bar{U}.$$

Observe that p is not unitless. $1/p$ has units of phase-space volume!

Definition (11.3): The Gibbs entropy for a system of N particles in phase space $Q = \Omega \times \mathbb{R}^3$ with particle density $f: Q \rightarrow [0, \infty)$ is defined as

$$(11.6) \quad S_G[f] := -k_B \iint_Q f(\vec{x}, \vec{v}) \log\left(\frac{f(\vec{x}, \vec{v})}{P_0 N}\right) d\vec{v} d\vec{x},$$

where $P_0 > 0$ is a normalization factor with the property that

$$\left[\frac{f(\vec{x}, \vec{v})}{P_0 N} \right] = \left[\frac{p(\vec{x}, \vec{v})}{P_0} \right] = 1.$$

The average entropy (per particles) is defined as

$$\bar{S}_G[f] = \frac{S_G[f]}{N}.$$

Thus,

$$(11.7) \quad \bar{S}_G[f] = -k_B \iint_Q p(\vec{x}, \vec{v}) \log\left(\frac{p(\vec{x}, \vec{v})}{P_0}\right) d\vec{v} d\vec{x}.$$

Remark: The arguments for a transcendental function, like $\log$, must be unitless.

As in the discrete (quantized) energy case the equilibrium state is the one that maximizes

the Gibbs entropy.

Theorem (11.1): The equilibrium particle density for non-interacting particles with constraints

$$(11.8) \quad N = \iint_Q f(\vec{x}, \vec{v}) d\vec{v} d\vec{x}$$

and

$$(11.9) \quad U = \iint_Q f(\vec{x}, \vec{v}) u(\vec{x}, \vec{v}) d\vec{v} d\vec{x}$$

is given by

$$(11.10) \quad f(\vec{x}, \vec{v}) = f^{\text{eq}}(\vec{x}, \vec{v}) = P_0 \frac{N}{Z} \exp\left(\frac{-\beta u(\vec{x}, \vec{v})}{k_B}\right),$$

where Z is the partition function

$$(11.11) \quad Z = P_0 \iint_Q \exp\left(\frac{-\beta u(\vec{x}, \vec{v})}{k_B}\right) d\vec{v} d\vec{x}$$

and β is a Lagrange multiplier that can be uniquely determined as a function of U and N .

Proof: let us minimize the functional

$$\begin{aligned} J[f] = & k_B \iint_Q f(\vec{q}) \log\left(\frac{f(\vec{q})}{P_0 N}\right) d\vec{q} \\ & + \alpha \left\{ \iint_Q f(\vec{q}) d\vec{q} - N \right\} + \beta \left\{ \iint_Q f(\vec{q}) u(\vec{q}) d\vec{q} - U \right\} \end{aligned}$$

where

$$\vec{q} := (\vec{x}, \vec{v}) \in Q$$

and $\alpha, \beta \in \mathbb{R}$ are Lagrange multipliers. To compute the variation observe that

$$\begin{aligned} \frac{d}{ds} [J(f+sg)] \Big|_{s=0} \\ &= \iint_Q g(\vec{q}) \left\{ k_B \log \left(\frac{f(\vec{q})}{NP_0} \right) + k_B + \alpha + \beta u(\vec{q}) \right\} d\vec{q} \\ &= 0 \end{aligned}$$

for any g with $f+sg: Q \rightarrow [0, \infty)$, et cetera.

$$k_B \log \left(\frac{f(\vec{q})}{NP_0} \right) + k_B + \alpha + \beta u(\vec{q}) = 0, \quad \forall \vec{q} \in Q.$$

Solving for f , we find

$$f^{\text{eq}}(\vec{q}) = P_0 \frac{N}{Z} \exp \left(\frac{-\beta u(\vec{q})}{k_B} \right),$$

where

$$Z = P_0 \iint_Q \exp \left(\frac{-\beta u(\vec{q})}{k_B} \right) d\vec{q}$$

is the unitless partition function. Here we have used the constraint (11.8) to eliminate α . Thus,

$$\iint_Q f^{\text{eq}}(\vec{q}) d\vec{q} = N.$$

We contend that, as before, the energy constraint (11.9) is sufficient to eliminate β , and in fact β is a unique positive function of V and N .

We will demonstrate this momentarily. //

Theorem (11.2): At equilibrium, the Gibbs entropy satisfies

$$(11.12) \quad S_{\text{eq}}(V, N) := S_G[f^{\text{eq}}] = \beta U + k_B \log(Z) N,$$

and at equilibrium the derivatives are

$$(11.13) \quad \frac{\partial S_{\text{eq}}}{\partial U} = \beta \quad \text{and} \quad \frac{\partial S_{\text{eq}}}{\partial N} = k_B \log(Z).$$

Proof: At equilibrium,

$$f^{\text{eq}}(\vec{q}) = P_0 \frac{N}{Z} \exp\left(\frac{-\beta U(\vec{q})}{k_B}\right)$$

where

$$Z = P_0 \iint_Q \exp\left(\frac{-\beta U(\vec{q})}{k_B}\right) d\vec{q}$$

$$S_{\text{eq}}(V, N) = S_G[f^{\text{eq}}]$$

$$= -k_B \iint_Q f^{\text{eq}}(\vec{x}, \vec{v}) \log\left(\frac{f^{\text{eq}}(\vec{x}, \vec{v})}{P_0 N}\right) d\vec{v} d\vec{x}$$

$$\begin{aligned}
&= -k_B \iint_Q \frac{P_0 N}{Z} \exp\left(\frac{-\beta u(\vec{q})}{k_B}\right) \log\left(\frac{\exp\left(\frac{-\beta u(\vec{q})}{k_B}\right)}{Z}\right) d\vec{q} \\
&= -\frac{k_B P_0 N}{Z} \iint_Q \exp\left(\frac{-\beta u(\vec{q})}{k_B}\right) \left\{ \frac{-\beta u(\vec{q})}{k_B} - \log(Z) \right\} d\vec{q} \\
&= \frac{P_0 N}{Z} \beta \iint_Q \exp\left(\frac{-\beta u(\vec{q})}{k_B}\right) u(\vec{q}) d\vec{q} \\
&\quad + \frac{k_B P_0 N}{Z} \log(Z) \iint_Q \exp\left(\frac{-\beta u(\vec{q})}{k_B}\right) d\vec{q} \\
&= \beta U + k_B N \log(Z).
\end{aligned}$$

Suppose that τ is a process parameter and assume that $(\vec{q}, \tau) \mapsto f(\vec{q}, \tau)$,

$$f : Q \times [0, \infty) \rightarrow [0, \infty).$$

Thus,

$$N(\tau) = \iint_Q f(\vec{q}, \tau) d\vec{q}.$$

It now follows that

$$\begin{aligned}
&\frac{d}{d\tau} S_G[f(\cdot, \tau)] \\
&= -k_B \frac{d}{d\tau} \iint_Q f(\vec{q}, \tau) \log\left(\frac{f(\vec{q}, \tau)}{P_0 N(\tau)}\right) d\vec{q}
\end{aligned}$$

$$= -k_B \iint_Q \left\{ \dot{f} \log \left(\frac{f}{P_0 N} \right) + f \left[\frac{\dot{f}}{f} - \frac{\dot{N}}{N} \right] \right\} d\vec{q}$$

$$= -k_B \iint_Q \dot{f} \left\{ \log \left(\frac{f}{P_0 N} \right) + 1 \right\} d\vec{q}$$

$$+ k_B \frac{\dot{N}}{N} \iint_Q f d\vec{q}$$

$$= -k_B \iint_Q \dot{f} \log \left(\frac{f}{P_0 N} \right) d\vec{q}.$$

Here, we have used the notation

$$\dot{f} = \frac{\partial f}{\partial \tau}, \quad \dot{N} = \frac{dN}{d\tau}.$$

Next, observe that

$$U(\tau) = \iint_Q f(\vec{q}, \tau) u(\vec{q}) d\vec{q},$$

so that

$$\frac{dU}{d\tau} = \dot{U} = \iint_Q \dot{f} u d\vec{q}.$$

Suppose that $(\vec{q}, \tau) \mapsto f(\vec{q}, \tau)$,

$$f: Q \times [0, \infty) \rightarrow [0, \infty),$$

is a path through number density space, with

$$f(\cdot, \tau_0) = f^{eq}(\cdot).$$

Then,

$$\text{and } N(\tau_0) = N, \quad U(\tau_0) = U,$$

$$\frac{d}{d\tau} S_G[f(\cdot, \tau_0)]$$

$$= -k_B \iint_Q \dot{f}(q, \tau_0) \log \left(\frac{f(\vec{q}, \tau_0)}{P_0 N(\tau_0)} \right) d\vec{q}$$

$$= -k_B \iint_Q \dot{f}(q, \tau_0) \log \left(\frac{f^{eq}(\vec{q})}{P_0 N} \right) d\vec{q}$$

$$= -k_B \iint_Q \dot{f} \left[\left(-\frac{\beta u(\vec{q})}{k_B} \right) - \log(Z) \right] d\vec{q}$$

$$= \beta \dot{U} + k_B \dot{N} \log(Z).$$

As in the proof of Theorem (10.2), we can conclude that

$$\frac{\partial S_{eq}}{\partial U} = \beta \quad \text{and} \quad \frac{\partial S_{eq}}{\partial N} = k_B \log(Z). //$$

Definition (11.3): The statistical temperature is defined by the expression

$$(11.14) \quad \frac{1}{T} := \frac{\partial S_{eq}}{\partial U},$$

where

$$S_{eq}(U, N) = S_q[f^*].$$

The statistical chemical potential is defined
via

(11.15)

$$\frac{\mu}{T} := \frac{\partial S_{eq}}{\partial N}.$$

Thus

(11.16)

$$T = \frac{1}{\beta}$$

and

(11.17)

$$\mu = k_B T \log(Z).$$